



Seminar

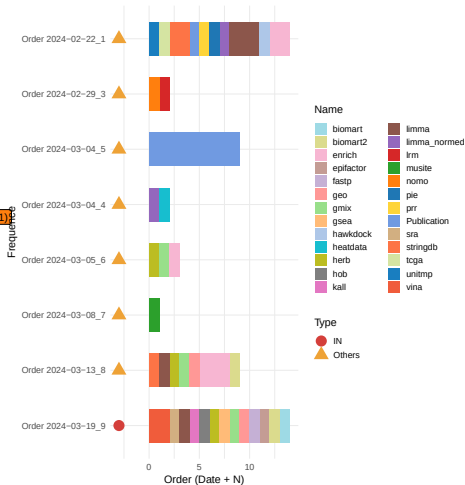
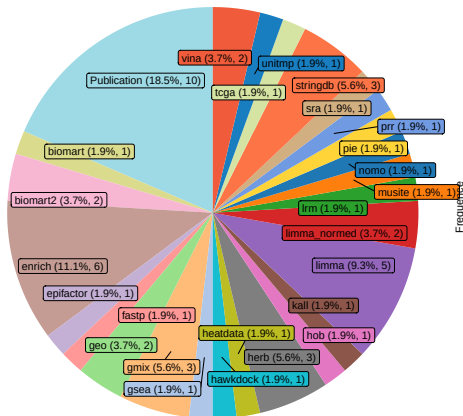
Lichuang Huang

Wie-Biotech

2024-04-12

Tasks summary

Overview 2024-03



New Methods

- unitmp: UNIfied database of TransMembrane Proteins
- prr: (RNA-seq) Clinical Chemotherapeutic Response
- musite (GPS-MSP, PRmePred): (MusiteDeep) Post-translational modification site
 - Web
 - Python
- epifactor: Epigenetic regulators

Article sharing

Concept

- WGS / WES
- GWAS
- Gene locus (基因座)
- Allele (等位基因)
- Linkage Disequilibrium (连锁不平衡)
- ...
- eQTL (GTEx database): SNPs $\Rightarrow$ mRNA expression levels

GWAS

- Disease $\Leftrightarrow$ Variants
- Variants $\Rightarrow$ Genes
- Variants $\Rightarrow \dots \Rightarrow$ Genes



- Variants $\Rightarrow ? \Rightarrow$ Disease

Article with multiple tools



Contents lists available at [ScienceDirect](#)

Behavioural Brain Research

journal homepage: www.elsevier.com/locate/bbr



Research report

Prediction of causal genes and gene expression analysis of attention-deficit hyperactivity disorder in the different brain region, a comprehensive integrative analysis of ADHD



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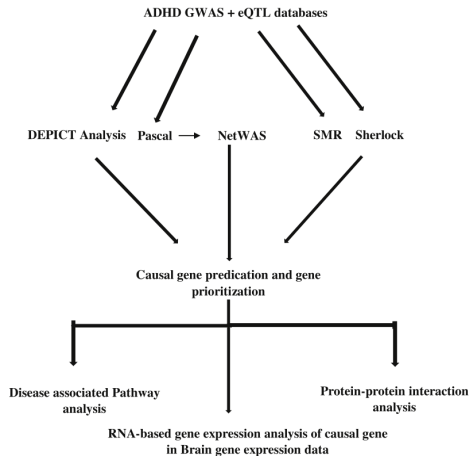
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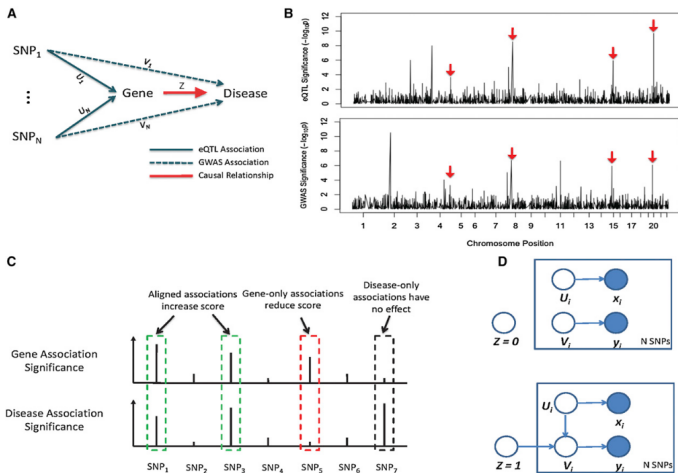
Flow Chart

- Sherlock
- SMR
- NetWAS (GIANT websevier)
- Depict

A. Fehra, et al.

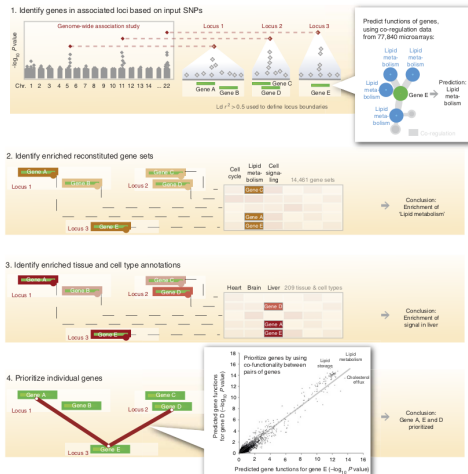


Sherlock (and SMR) via eQTL



Depict via Prediction

- 1 预测关联位点对应基因的功能
- 2 候选基因功能富集分析
- 3 候选基因表达特异性分析 (细胞、组织)
- 4 Gene prioritization



Schedule

Schedule

- Finish orders
- New methods

Thanks